Digitalization of business (including customer) processes to enhance operational efficiencies and generating error free billing

Mr. Amit Arya, Mr. Saurabh Kumar, Mr. Shubham Tyagi

AGM, Tata Power Delhi Distribution Limited, Delhi, India
Manger, Tata Power Delhi Distribution Limited, Delhi, India
Manger, Tata Power Delhi Distribution Limited, Delhi, India

Keywords: GIS, ADMS, Digital Transformation, CRM, System Interlockings, Change Management, Digitalization, Digital Transformation, TQM

Abstract

Digitalization in the power utility sector enhances customer service, streamlines operations, and lowers costs. Customers play a critical role in power distribution utilities. Power is procured from generators and distributed to a wide range of customers, including domestic, commercial, and industrial users. The revenue generation for power utilities is dependent on customers paying their bills after the completion of the billing cycle.

This paper discusses the digitalization of business processes to enhance operational efficiencies and generate error-free billing. It outlines the six key steps involved in the customer lifecycle, from applying for a new connection to the payment of bills, and highlights the role of technologies such as smart meters, Customer Relationship Management (CRM) systems, and data-driven decision-making. Findings show significant improvements in connection energization timelines, billing accuracy, and customer satisfaction.

This document focuses on Tata Power DDL's complete digital transformation journey, with CRM as a foundational principle. The paper touches upon aspects related but not limited to Data capturing mechanisms, Data processing techniques, Process, and System interlockings to ensure enterprise level data flow and utilization, Change Management and Measurement and Control techniques for Data Management and Quality improvement using Total Quality Management (TQM) Principles.

1 Introduction

The power distribution sector is a critical infrastructure that forms the backbone of modern economies. In this context, digitalization serves as a transformative enabler for operational efficiency, customer satisfaction, and cost reduction. Traditional manual processes in utilities have been characterized by inefficiencies, billing errors, and delayed services, which affect both customer experience and revenue generation.

This paper aims to illustrate the impact of digitalization in addressing these challenges and creating a robust system for seamless utility operations. By focusing on Tata Power Delhi Distribution Limited (TPDDL) as a case study, we demonstrate how digital solutions can drive efficiency gains, improve service quality, and ensure customer-centric operations.

1.1 The Digitalization Journey in Power Utilities

1.1.1 Key Steps in the Customer Lifecycle

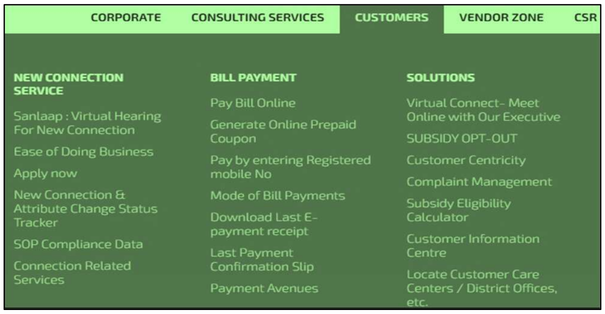
The digitalization of customer-centric processes in power utilities can be divided into six major steps:

Submitting an Online Application for a New Connection
Customers can apply for new connections through an intuitive online portal, eliminating the need for physical visits and paperwork.

This step ensures transparency and reduces application processing times.

- Document Processing, Validation, and Technical Feasibility Assessment
- Digital tools validate customer-provided documents and assess technical feasibility for connection.

https://doi.org/10.1049/icp.2025.2422

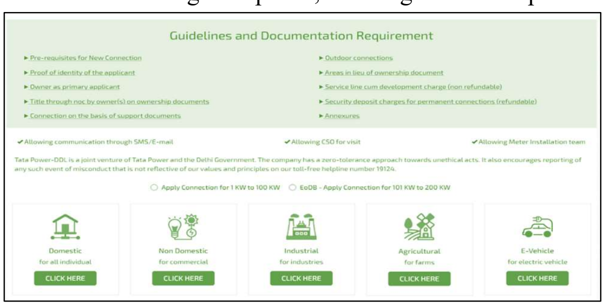


1.1 Figure Captions: Web Portal

Automated workflows reduce manual intervention, increasing accuracy and speed.

Providing Technical Feasibility via an Online Portal

Once feasibility is determined, the results are communicated to customers through the portal, ensuring real-time updates.



1.2 Figure Captions: New Connection Processing

This feature improves customer trust and engagement.

Improving the Ease of Access to Electricity Connections in India

The **World Bank** ranks 190 economies globally on various parameters, including **"Getting Electricity,"** which evaluates the ease of acquiring electricity connections. The ranking is based on four key factors:

1. **Time taken to get an electricity connection.**
2. **Cost associated with obtaining a new connection.**
3. **Procedures required for the connection process.**
4. **Reliability of the electricity supply.**

For a 140KW / 150KVA connection on a 415V supply, these factors significantly impact the customer experience.

India's ranking for **"Getting Electricity"** has seen remarkable improvement. From **137th place in 2015**, India advanced to **22nd rank in 2019**, thanks to the concerted efforts of key stakeholders, including:

- The **Ministry of Power**
- **Central Electricity Authority (CEA)**
- State governments, particularly **Delhi** and **Maharashtra**
- Distribution companies like **Tata Power-DDL** and **BEST**

2 Key Reforms Contributing to Improved Rankings

Several reforms implemented across Delhi and India have contributed to this improvement:

1. **Central Electricity Authority (CEA) Initiatives**

Installation of 500KVA transformers in Pole Mounted Substations (up from 250KVA).
Empowering DISCOMs with Electrical Inspection Clearance for installations up to 11KV.

2. **Delhi Electricity Regulatory Commission (DERC) Measures**

Introduction of a 7-day energization timeline for new connections, provided the connection is feasible from the existing distribution network, and no road cutting or Right of Way (RoW) permission is required.

Simplified documentation for obtaining new connections (only ID proof and ownership proof required).

Rationalization of tariffs for 415V connections up to 150KVA.

Expansion of supply capacity to 200KW/215KVA on LT loads.

3. **Delhi Government Initiatives**

Online approval mechanism for Right of Way (RoW) from PWD, streamlining the approval process.

Introduction of an online portal for new electricity connection applications and document submissions.

Demand Note charges for new connections are now added to the applicant's first electricity bill, simplifying the billing process.

EODB (Ease of Doing Business) initiatives include an online submission process for applications and a streamlined field inspection and energization process.

The ability for customers to apply for new connections ranging from 101KW to 200KW at LT supply.

Upgraded SCADA system to an Advanced Distribution Management System (ADMS), integrating SCADA, DMS, OMS, GIS, ERP, CRM, FFA, and MDMS. This integrated platform enhances operational efficiency, improves response times, and ensures real-time monitoring.

These reforms have significantly reduced the **time** and **complexity** involved in securing a new connection, enhancing overall **service delivery** and **customer satisfaction**.

2 Connecting and Installing a Smart Meter at the Customer's Site

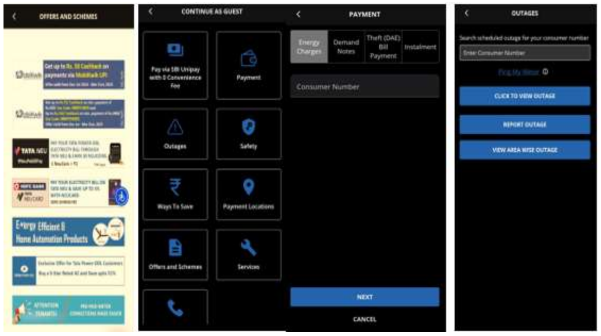
Smart meters play a vital role in the digitalization process, enabling real-time energy monitoring and reducing meter-reading errors.
Installation is tracked digitally to ensure SLA adherence.

2.1 Figure Captions: Getting Connection: EODB

Monthly Bill Generation by the Utilities

Advanced CRM systems like SAP generate accurate monthly bills based on real-time meter data.

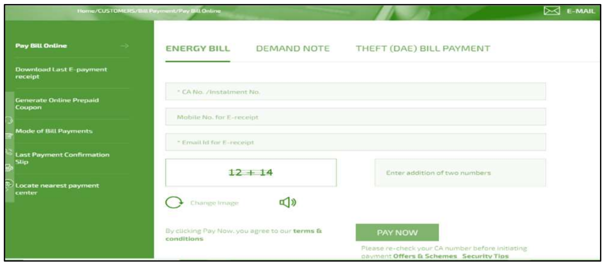
Automation minimizes errors and administrative costs.



2.2 Figure Captions: Online Bill Generation

Customer Payment of the Monthly Bill

Digital payment gateways offer multiple options for seamless customer payments, enhancing convenience and timeliness.



2.3 Figure Captions: Bill Payment

Key Enablers of Digitalization

Smart Meters

Smart meters are a cornerstone of digital transformation in power utilities. They enable:

- Accurate, real-time energy consumption monitoring.
- Detection of energy theft and system losses.
- Remote meter reading, eliminating the need for manual visits.

Customer Relationship Management (CRM)

A CRM system, such as SAP, integrates customer data and automates billing processes. Key features include:

- Real-time billing.
- Complaint tracking and resolution.
- Personalized customer interactions.

Data-Driven Decision-Making

By leveraging analytics and visualization tools, utilities can:
Identify process bottlenecks and inefficiencies.
- Monitor Key Performance Indicators (KPIs) for performance tracking.
- Enable proactive maintenance and service delivery.

Integration of SLA, TQM, IMS, and KPI Monitoring

Service Level Agreements (SLA):

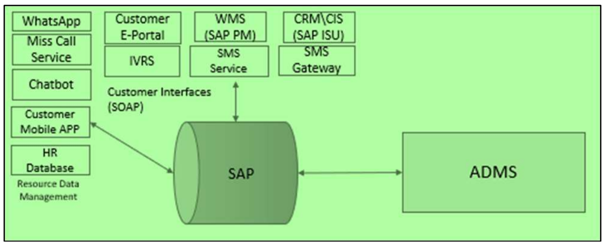
- SLA adherence ensures timely connection energization and complaint resolution.

Total Quality Management (TQM):

- TQM principles enhance the quality of processes and outputs.

Integrated Management Systems (IMS):

- IMS frameworks consolidate multiple processes into a unified approach for better control.



2.4 Figure Captions: ADMS & SAP data flow

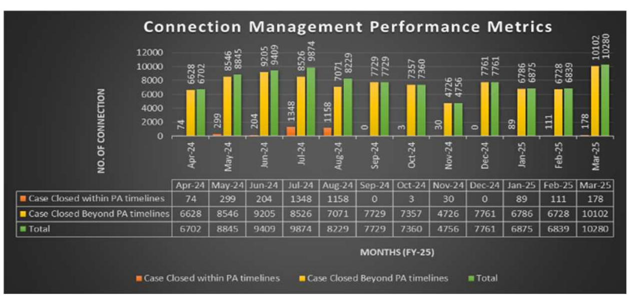
Key Performance Indicators (KPIs):

- Metrics like billing efficiency, complaint resolution time, and connection energization cycles measure the impact of digitalization.

3 Results and Success Stories

Connection Energization Cycle

- Digitalization has reduced the average connection energization cycle to 7 days, compared to industry averages of 15-20 days.



	Apr-24	May-24	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24	Jan-25	Feb-25	Mar-25
Case Closed within PA timelines	74	299	204	1348	1158	0	3	30	0	89	111	178
Case Closed Beyond PA timelines	6628	8546	9205	8526	7071	7729	7357	4726	7761	6786	6728	10102
Total	6702	8845	9409	9874	8229	7729	7360	4756	7761	6875	6839	10280

3.1 Figure Captions: Connection Management Performance Metrics

- Real-time tracking ensures adherence to timelines and customer satisfaction.

Billing Efficiency

- CRM-based systems have achieved 99.9% billing accuracy, significantly reducing revenue leakage and customer complaints.
- Automated meter readings and error-free billing processes contribute to this improvement.

Reduction in Billing Complaints

- Customer complaints related to billing errors have decreased by 50%, reflecting the reliability of digital systems.

Enhanced Operational Efficiency

- Streamlined operations have led to a 50% improvement in overall efficiency, reducing administrative overheads and energy losses.

Performance Assurance (PA) Timelines: A Key KPI for Regulatory Compliance in the Power Sector

In the power utility sector, maintaining **Performance Assurance (PA) timelines** is a critical Key Performance Indicator (KPI) for ensuring operational excellence and regulatory compliance. Regulatory bodies, such as the **Regulatory Commission**, have established strict timelines for various aspects of utility operations, which must be adhered to by distribution companies. These timelines are set to guarantee a high standard of service delivery and accountability, ensuring that utilities meet customer expectations while maintaining efficiency and reliability in the system.

Importance of PA Timelines

1. **Regulatory Compliance**: Adhering to PA timelines is mandatory for utilities as per regulatory guidelines. Failure to comply may result in penalties or adverse regulatory actions, undermining the trust between utilities and regulatory bodies. For example, ensuring that **new connections** are energized within a stipulated number of days, such as the **7-day energization rule** for new applications, ensures that customers experience minimal delays in receiving service.
2. **Customer Satisfaction**: Timely adherence to PA timelines directly impacts customer experience. Quick connection setups, bill generation, and dispute resolution foster positive customer relationships. Utilities that meet or exceed these expectations are more likely to build customer trust, reduce complaints, and retain customers.
3. **Operational Efficiency**: PA timelines drive internal operational efficiency by setting clear benchmarks for teams. By implementing digital platforms such as **Advanced Distribution Management Systems (ADMS)** and **CRM** tools, utilities can monitor progress in real-time, track performance, and ensure timely execution of tasks. This proactive approach reduces delays, enhances resource allocation, and improves overall system reliability.
4. **Data-Driven Decision-Making**: Maintaining and tracking PA timelines requires accurate data management and performance tracking. Real-time monitoring tools allow decision-makers to identify bottlenecks and inefficiencies quickly. By using data analytics, utilities can optimize their operations, enhance their response times, and address issues before they impact customers.

5. **Financial Implications**: Meeting PA timelines also has financial benefits. Efficient processes, coupled with timely execution, reduce operational costs by minimizing the need for rework and administrative overhead. Moreover, by complying with regulatory deadlines, utilities avoid penalties, thereby protecting their financial standing.

Challenges and Mitigation Strategies Improve PA Timeline Compliance

Data Integration: Integrating legacy systems with modern digital tools can be complex.

Customer Adaptation: Encouraging customers to adopt digital platforms requires awareness campaigns.

Cybersecurity Risks: Digitalization increases the risk of data breaches and cyber-attack

Mitigation Strategies

Robust IT Infrastructure: Ensuring scalability and resilience in IT systems.

Customer Education: Conducting workshops and campaigns to familiarize customers with digital tools.

Cybersecurity Measures: Implementing advanced security protocols and regular audits.

Strategies to Improve PA Timeline Compliance

1. **Digitalization of Processes**: By incorporating automated systems such as **GIS**, **SCADA**, and **MDMS**, utilities can streamline various operational tasks, from meter readings to billing, reducing manual errors and accelerating service delivery. Digital platforms allow real-time tracking and better coordination among teams, ensuring timely completion of activities.
2. **Regular Performance Reviews**: Conducting periodic reviews of PA timelines ensures that performance remains on track. Regular audits and reports provide insights into areas where timelines are being missed and allow for corrective action before further delays occur.
3. **Training and Capacity Building**: Ensuring that employees are well-trained and equipped to handle tasks efficiently is vital. By promoting a culture of continuous improvement and providing training on best practices, utilities can improve adherence to timelines and meet the regulatory expectations.
4. **Collaboration and Coordination**: Effective communication and coordination between departments are essential to meet PA timelines. Utilities should foster cross-departmental collaboration to ensure that all necessary steps are completed in a timely manner, reducing delays caused by inter-departmental bottlenecks.

4 Conclusion

The digitalization of power utilities represents a paradigm shift in how services are delivered and managed. By automating processes, enhancing operational efficiencies, and prioritizing customer-centricity, utilities can create a sustainable and future-ready framework.

The concerted efforts by the Ministry of Power, Central Electricity Authority, Delhi Government, and utilities have significantly transformed the process of obtaining electricity connections in India. Key reforms, including simplified documentation, reduced energization timelines, and digital platforms for applications, have improved operational efficiency and customer satisfaction. Tata Power-DDL's implementation of an Advanced Distribution Management System (ADMS) further enhances service delivery. These initiatives have not only streamlined procedures but also propelled India's ranking in "Getting Electricity" from 137th in 2015 to 22nd in 2019, showcasing the success of these digitalization and reform efforts in the power sector.

In the power utility sector, adhering to Performance Assurance (PA) timelines is a key factor in ensuring both regulatory compliance and customer satisfaction. By leveraging digital tools, optimizing processes, and fostering a culture of accountability, utilities can meet these stringent timelines, resulting in enhanced operational performance, improved customer experiences, and regulatory adherence.

Currently organization is committed in achieving a 7-day connection energization cycle, 99.9% billing efficiency, and a 50% reduction in billing complaints underscores the transformative potential of digitalization. As emerging technologies continue to evolve, power utilities must leverage these advancements to meet the growing demands of their customers and stakeholders.

5 Acknowledgements

We would like to extend our gratitude to Tata Power Delhi Distribution Limited for providing the necessary support, resources, and infrastructure that facilitated this study. The invaluable guidance and expertise of the leadership and team members at Tata Power-DDL have been instrumental in shaping this research and driving its successful completion.

We are especially thankful for the access to data, advanced systems, and tools such as the ADMS, GIS, and CRM platforms, which were crucial in analysing and demonstrating the impact of digitalization in the power utility sector.

Finally, the encouragement and collaborative spirit of the Tata Power-DDL team played a vital role in ensuring the quality and relevance of this work.

6 References

1. Yukihiro Ando, Pankaj Kumar (2021) ‘Daily Management the TQM Way’.
2. Pavan Soni (2020) ‘Design your Thinking’.
3. Internal Data from PA group: Not for circulation.

Websites

1. https://powermin.gov.in/sites/default/files/uploads/Ease_of_doing_business_getting_electricity.pdf
2. https://www.tatapowerddl.com/billpay/demandnote/demandnoteportal.asp